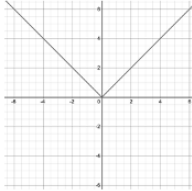


Algebra 1
Unit 1
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

1. The equation $y = \sqrt[3]{x}$ is the parent function for

- ☐ Linear function
- ☐ Quadratic function
- ☐ Cubic function
- ☐ Square root function
- ☒ Cube root function
- ☐ Absolute Value function
- ☐ Exponential function

2. The graph  is the parent function for

- ☐ Linear function
- ☐ Quadratic function
- ☐ Cubic function
- ☐ Square root function
- ☐ Cube root function
- ☒ Absolute Value function
- ☐ Exponential function

3. The equation $y = x^2$ is the parent function for

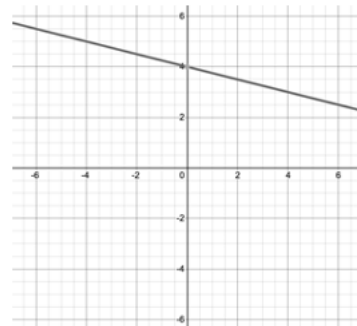
- ☐ Linear function
- ☒ Quadratic function
- ☐ Cubic function
- ☐ Square root function
- ☐ Cube root function
- ☐ Absolute Value function
- ☐ Exponential function

For equations 4-10, classify the function family for each equation or graph.

4. $y = 3^{4x}$

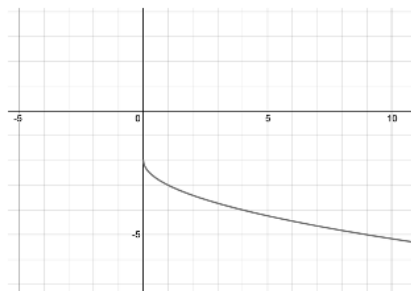
Exponential Function

5.



Linear Function

6.

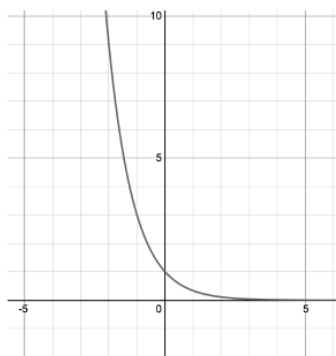


Square Root Function

7. $y = x^3 + 5$

Cubic Function

8.

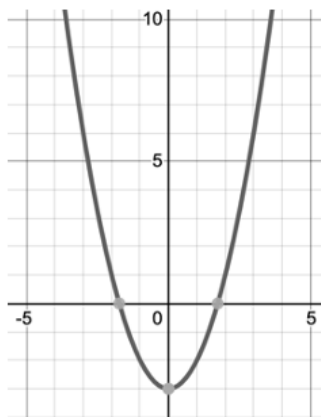


Exponential Function

9. $y = -|x + 3|$

Absolute Value Function

10.



Quadratic Function

Algebra 1

Unit 1

Lesson 2 Homework

Name **Answer Key** _____

Date _____

Period _____

Given the function $r(x) = -\frac{2}{3}x + 4$, find each value.

$$1. \quad r(-6) = -\frac{2}{3}(-6) + 4$$

$$= \frac{12}{3} + 4 = 4 + 4 = \boxed{8}$$

$$2. \quad r\left(\frac{1}{2}\right) = -\frac{2}{3}\left(\frac{1}{2}\right) + 4$$

$$= -\frac{1}{3} + 4 = \boxed{\frac{11}{3}}$$

3. For what value of x is $r(x) = -4$?

$$3 \left[-4 = -\frac{2}{3}x + 4 \right] 3$$

$$-12 = -2x + 12$$

$$\frac{-12 \quad -12}{-24 = -2x}$$

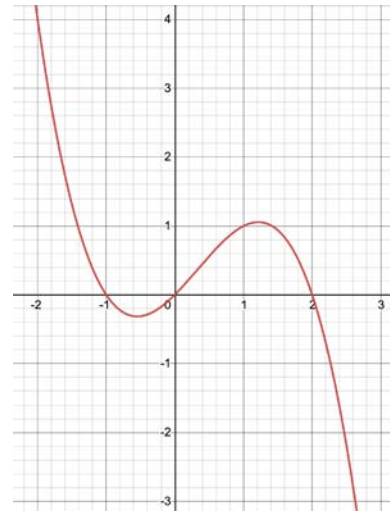
$$-24 = -2x$$

$$\boxed{x = 12}$$

Use the graph of $y = f(x)$ to find each value.

$$4. \quad f(-2) = 4$$

$$5. \quad f(1) = 1$$



6. For what value of x is $f(x) = 0$?

$$x = 0$$

Use the table of values representing $m(x)$ to find each value.

x	$m(x)$
-6	4.5
8	2
0	-3.2
-24	-13
7	-9
3	0

7. $m(8) = 2$

8. $m(-24) = -13$

9. What is the y -intercept of the function in the table?
 $(0, -3.2)$

10. Given $d(x) = -2|x^2 - 3.6|$, find the value for $x = 5.5$.

$$\begin{aligned}d(5.5) &= -2|(5.5)^2 - 3.6| \\&= -2|30.25 - 3.6| \\&= -2|26.65|\end{aligned}$$

$$d(5.5) = -53.3$$

The function $C(t) = 25.50t - 16.75$ represents the cost of buying a certain number of tickets to the concert with a \$16.75 coupon. The cost, C , is measured in dollars for t tickets.

- Find the value of $C(4)$.
 $c(4) = 25.50(4) - 16.75$
 $= 102 - 16.75 = \$82.25$
- What does $C(4)$ represent in the context of the problem?
The cost for 4 tickets.
- Is the value of $C(6.5)$ reasonable to interpret in the context of the problem?
Not reasonable for there's no half (0.5) of a ticket.

A function $T(h)$ is modeled in the graph below. It shows the temperature in Miami, Florida, where T is the temperature and h is the number of hours since 7 A.M.

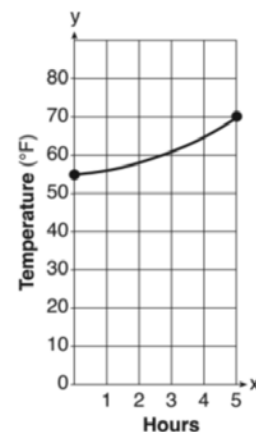
- Find the approximate value of the graph for $T(1)$.

$$T(1) \approx 56^{\circ}\text{F}$$

- What does $T(1)$ mean in the context of the problem?
The temperature during 8 A.M.

- At what time is the temperature 70°F ?
12 noon

- Why does it make sense that $T(4) > T(2)$ in this context?
Temperature is higher at 11 A.M. than 9 A.M.



The table below represents the number of new homes available in Broward County and the sale prices, in thousands of dollars in 2021.

Sale Price, p (in thousands of dollars)	200	240	280	320	360	400
Number of New Homes Available, $n(p)$	112	105	87	62	31	25

8. Explain the meaning of $n(0) = 0$.
There is a zero sale price because there are no new homes available.
9. Determine the value of $n(320)$.
62 homes
10. Interpret $n(320)$ for the given context.
There will be 62 new homes available if the sale price is \$320,000.

Algebra 1

Unit 1

Lesson 4 Homework

Name **Answer Key** _____

Date _____

Period _____

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $(4.5^2)^5 = 4.5^{10}$

2. $\left(\frac{3}{7}\right)^{-3} = \frac{7^3}{3^3}$

3. $(-2 \cdot 6)^4 = (-2)^4(6)^4$

4. $\frac{(-3.1)^{10}}{(-3.1)^3} = (-3.1)^7$

5. $7^{-4} \cdot 7 \cdot 7^3 = 7^0 = 1$

Find the value of each for questions 6-7.

6. $\sqrt[3]{64} = 64^{\frac{1}{3}} = (4^3)^{\frac{1}{3}} = 4$

7. $\sqrt{64} = 64^{\frac{1}{2}} = (8^2)^{\frac{1}{2}} = 8$

8. Explain why the value of the $\sqrt{64}$ is different from the value of the $\sqrt[3]{64}$.
Different index $\rightarrow$ Different exponents

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{y} \cdot \sqrt{y} = 15$

$$\sqrt{y^2} = 15$$

$$y = 15$$

$$y = \underline{\quad 15 \quad}$$

10. $\sqrt[3]{g} \cdot \sqrt[3]{g} \cdot \sqrt[3]{g} = -3$

$$\sqrt[3]{g^3} = -3$$

$$g = \underline{\quad -3 \quad}$$

Algebra 1

Unit 1

Lesson 5 Homework

Name **Answer Key** _____

Date _____

Period _____

1. In radical form, the expression $x^{\frac{1}{2}}$ is equivalent to $\sqrt{x}$.

2. In exponential form, the expression $\sqrt[3]{x}$ is equivalent to $x^{\frac{1}{3}}$.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-125)^{\frac{1}{3}} = \sqrt[3]{-125} = -5$

4. $(1024)^{\frac{1}{5}} = \sqrt[5]{1024} = 4$

5. $49^{\frac{1}{2}} = \sqrt{49} = 7$

6. Simplify the expression using the laws of exponents and then find the value.

$$36^{-\frac{5}{4}} \cdot 36^{\frac{7}{4}} = 36^{\frac{1}{2}} = \sqrt{36} = 6$$

The expression $\frac{81^{\frac{10}{12}}}{81^{\frac{7}{12}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?
Quotient of powers

8. Rewrite the expression as an equivalent expression.

$$81^{\frac{3}{12}} = 81^{\frac{1}{4}}$$

9. Write simplified expression in radical form.

$$\sqrt[4]{81}$$

10. Evaluate the radical expression.

$$\sqrt[4]{81} = 3$$

Algebra 1

Unit 1

Lesson 6 Homework

Name **Answer Key** _____

Date _____

Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(-8^{\frac{1}{3}}\right)^2$ and $(-8^2)^{\frac{1}{3}}$.

$$\left(-8^{\frac{1}{3}}\right)^2 = -8^{\frac{1}{3} \cdot 2} = -8^{\frac{2}{3}} = -8^{2 \cdot \frac{1}{3}} = (-8^2)^{\frac{1}{3}}$$

2. Identify all of the expressions equivalent to $\left(27^{\frac{2}{3}}\right)^2$.

- $27^{\frac{3}{4}}$
- $\sqrt[4]{27^3}$
- $(27^2)^{\frac{2}{3}}$ ✓
- $\sqrt[3]{27^4}$ ✓
- $(\sqrt[3]{27})^4$ ✓

For questions 3-8, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.	$25^{\frac{5}{2}}$	$\left(25^{\frac{1}{2}}\right)^5$	$(25^5)^{\frac{1}{2}}$	$(\sqrt{25})^5$	$\sqrt{25^5}$
4.	$(-27)^{\frac{2}{3}}$	$\left[(-27)^{\frac{1}{3}}\right]^2$	$[(-27)^2]^{\frac{1}{3}}$	$(\sqrt[3]{-27})^2$	$\sqrt[3]{(-27)^2}$
5.	$70^{\frac{4}{5}}$	$\left(70^{\frac{1}{5}}\right)^4$	$(70^4)^{\frac{1}{5}}$	$(\sqrt[5]{70})^4$	$\sqrt[5]{70^4}$
6.	$8^{\frac{7}{3}}$	$\left(8^{\frac{1}{3}}\right)^7$	$(8^7)^{\frac{1}{3}}$	$(\sqrt[3]{8})^7$	$\sqrt[3]{8^7}$
7.	$121^{\frac{3}{2}}$	$\left(121^{\frac{1}{2}}\right)^3$	$(121^3)^{\frac{1}{2}}$	$(\sqrt{121})^3$	$\sqrt{121^3}$

For questions 8-10, rewrite each expression using a rational exponent in the form $a^{\frac{b}{c}}$.

8. $\sqrt[3]{(-48)^8}$

$$(-48)^{\frac{8}{3}}$$

9. $(\sqrt{18})^3$

$$18^{\frac{3}{2}}$$

10. $\frac{1}{\sqrt[4]{10^5}}$

$$\frac{1}{10^{\frac{5}{4}}} = 10^{-\frac{5}{4}}$$

Algebra 1
Unit 1
Lesson 7 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Write the expression $64^{\frac{3}{2}}$ in two different equivalent forms.

Expression 1	Expression 2
$(64^3)^{\frac{1}{2}}$	$\sqrt{64^3}$

2. The value of $64^{\frac{3}{2}}$ is

- ☐ 16
- ☐ 42
- ☐ 83
- ☐ 512

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-216)^{\frac{2}{3}} = \sqrt[3]{(-216)^2} = 36$

4. $-\left(\frac{32}{243}\right)^{\frac{4}{5}} = -\sqrt[5]{\left(\frac{32}{243}\right)^4} = -\frac{16}{81}$

5. $(144)^{-\frac{1}{2}} = \frac{1}{\sqrt{144}} = \frac{1}{12}$

Evaluate each expression in questions 6-10 without a calculator.

$$6. \quad -(2^{-2}) \left(81^{-\frac{1}{2}} \right) = - \left(\frac{1}{2^2} \right) \left(\frac{1}{\sqrt{81}} \right) = - \left(\frac{1}{4} \right) \left(\frac{1}{9} \right) = \frac{-1}{36}$$

$$7. \quad \left(32^{\frac{1}{5}} \right)^3 \cdot 8^{-1} = 2^3 \left(\frac{1}{8} \right) = \frac{8}{8} = 1$$

$$8. \quad - \left(\frac{16}{64} \right)^{-\frac{3}{2}} = - \left(\frac{64}{16} \right)^{\frac{3}{2}} = - \sqrt{\left(\frac{64}{16} \right)^3} = - \sqrt{4^3} = -8$$

$$9. \quad (-1^{10}) \cdot \left(-125^{\frac{1}{3}} \right) = (1)(-5) = -5$$

$$10. \quad \left(\frac{7}{21} \right)^{-3} \cdot (3^{-2}) = \left(\frac{21}{7} \right)^3 \cdot \left(\frac{1}{3^2} \right) = (27) \left(\frac{1}{9} \right) = 3$$

Algebra 1
Unit 1
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

1. Which of the following is not a radical expression that is equivalent to $\sqrt{108}$?

- A. $\sqrt{2} \cdot \sqrt{54}$
- B. $\sqrt{3} \cdot \sqrt{36}$
- ☒ C. $\sqrt{5} \cdot \sqrt{21}$
- D. $\sqrt{6} \cdot \sqrt{18}$

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

B. $\sqrt{3} \cdot \sqrt{36}$

Because $\sqrt{36}$ is 6 which is an integer.

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

3. $\sqrt[3]{200} = \sqrt[3]{2} \cdot \sqrt[3]{100}$

$\sqrt[3]{200} = \sqrt[3]{8} \cdot \sqrt[3]{25} = 2\sqrt[3]{25}$

4. $\sqrt{72} = \sqrt{9} \cdot \sqrt{8} = 3\sqrt{8}$

$\sqrt{72} = \sqrt{36} \cdot \sqrt{2}$

5. Rewrite $\sqrt[3]{200}$ as an integer multiplied by a radical.

$2\sqrt[3]{25}$

6. Rewrite $\sqrt{72}$ as an integer multiplied by a radical.

$6\sqrt{2}$

7. Identify all expressions that are equivalent to $-4\sqrt[3]{48}$.

- $2\sqrt[3]{6}$
- $-8\sqrt[3]{6}$ ✓
- $-8\sqrt[3]{12}$
- $32\sqrt[3]{2}$
- $-4 \cdot \sqrt[3]{8} \cdot \sqrt[3]{6}$ ✓
- $-4 \cdot \sqrt[3]{12} \cdot \sqrt[3]{4}$ ✓
- $-4 \cdot (24 \cdot 2)^{\frac{1}{3}}$ ✓

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $\sqrt{392} = \sqrt{196 \cdot 2} = \boxed{14\sqrt{2}}$

9. $\sqrt[3]{135} = \sqrt[3]{27 \cdot 5} = \boxed{3\sqrt[3]{5}}$

10. $-3\sqrt[3]{625} = -3\sqrt[3]{125 \cdot 5} = -3(5\sqrt[3]{5})$
 $= \boxed{-15\sqrt[3]{5}}$

Algebra 1
Unit 1
Lesson 9 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Which of the following expressions can be combined?

- A. $-3\sqrt{25} - 25\sqrt{3}$ $2\sqrt{4 \cdot 3} - 6\sqrt{9 \cdot 3}$
 B. $\sqrt{7} + 4\sqrt{14}$ $2 \cdot 2\sqrt{3} - 6 \cdot 3\sqrt{3}$
 C. $2\sqrt{12} - 6\sqrt{27}$ $4\sqrt{3} - 18\sqrt{3}$
 D. $5\sqrt{25} + \sqrt{20}$ $-14\sqrt{3}$

Three students simplified the expression $6\sqrt{27} - 5\sqrt{12}$ below. Their work is shown below.

Marc	Jess	Tamara
$6\sqrt{27} - 5\sqrt{12}$ $1\sqrt{13}$ $\sqrt{13}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $54\sqrt{3} - 20\sqrt{3}$ $34\sqrt{3}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $18\sqrt{3} - 10\sqrt{3}$ $8\sqrt{3}$

2. Which student simplified the expression correctly?

Tamara simplified the expression correctly.

3. Write a note to one of the students who made an error explaining their mistake.

Answers will vary.

Mark; first, in order to combine radical expressions they must be like terms, and these expressions are not. Secondly, you cannot add radicands together due to the order of operations.

4. Write a note to the other student who made an error explaining their mistake.

Answers will vary.

Jess, you did the first step correctly. However, when you simplified the radicands you forgot to take the principal square root of the numbers.

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

5. $-2\sqrt{18} + \sqrt{50}$
 $-2\sqrt{9 \cdot 2} + \sqrt{25 \cdot 2}$
 $-2 \cdot 3\sqrt{2} + 5\sqrt{2}$
 $-6\sqrt{2} + 5\sqrt{2}$
 $-\sqrt{2}$

6. $\sqrt[3]{56} - \sqrt[3]{189}$
 $\sqrt[3]{8 \cdot 7} - \sqrt[3]{27 \cdot 7}$
 $2\sqrt[3]{7} - 3\sqrt[3]{7}$
 $-\sqrt[3]{7}$

7. $6\sqrt{8} - 8\sqrt{6}$
 $6\sqrt{4 \cdot 2} - 8\sqrt{6}$
 $6 \cdot 2\sqrt{2} - 8\sqrt{6}$
 $12\sqrt{2} - 8\sqrt{6}$

These cannot be combined because they are unlike terms.

8. $200^{\frac{1}{2}} - 1000^{\frac{1}{3}}$
 $\sqrt[2]{100 \cdot 2} - \sqrt[3]{1000}$
 $10\sqrt{2} - 10$

These cannot be combined because they are not like terms.

9. $-\sqrt{80} + 6\sqrt{5} - 3\sqrt{20}$
 $-\sqrt{16 \cdot 5} + 6\sqrt{5} - 3\sqrt{4 \cdot 5}$
 $-4\sqrt{5} + 6\sqrt{5} - 3 \cdot 2\sqrt{5}$
 $2\sqrt{5} - 6\sqrt{5}$
 $-4\sqrt{5}$

10. $\sqrt[3]{250} - 4\sqrt[3]{54}$
 $\sqrt[3]{125 \cdot 2} - 4\sqrt[3]{27 \cdot 2}$
 $5\sqrt[3]{2} - 4 \cdot 3\sqrt[3]{2}$
 $5\sqrt[3]{2} - 12\sqrt[3]{2}$
 $-7\sqrt[3]{2}$

1. To multiply two radicals with the same index, multiply the coefficients together and then multiply the radicands together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

$$2. \quad \sqrt{13} \cdot \sqrt{13} = \sqrt{169} = \boxed{13}$$

$$3. \quad \frac{\sqrt{5}}{4\sqrt{125}} = \frac{\sqrt{5}}{4\sqrt{25 \cdot 5}} = \frac{\sqrt{5}}{20\sqrt{5}} = \boxed{\frac{1}{20}}$$

$$4. \quad \sqrt[3]{6} \cdot \sqrt[3]{9} = \sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = \boxed{3\sqrt[3]{2}}$$

$$5. \quad \sqrt{24} \cdot \sqrt{8} = \sqrt{192} = \sqrt{64 \cdot 3} = \boxed{8\sqrt{3}}$$

$$6. \quad \frac{\sqrt[3]{375}}{\sqrt[3]{648}} = \sqrt[3]{\frac{375}{648}} = \sqrt[3]{\frac{125}{216}} = \boxed{\frac{5}{6}}$$

$$7. \quad \sqrt[3]{18} \cdot \sqrt[3]{60} = \sqrt[3]{1080} = \sqrt[3]{216 \cdot 5} = \boxed{6\sqrt[3]{5}}$$

8. Identify all equivalent expressions to $\sqrt{15} \cdot \sqrt{6}$.

- ☐ $9\sqrt{10}$
- ☒ $\sqrt{90}$ ✓
- ☒ $(10 \cdot 9)^{\frac{1}{2}}$ ✓
- ☐ $3\sqrt{15}$
- ☒ $3\sqrt{10}$ ✓
- ☐ $15^{\frac{1}{3}} \cdot 6^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{2\sqrt{6}}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{30}}{5}$

10. $\frac{\sqrt{28}}{3\sqrt{8}} \cdot \frac{\sqrt{8}}{\sqrt{8}} = \frac{\sqrt{224}}{3 \cdot 8} = \frac{\sqrt{16 \cdot 14}}{24} = \frac{4\sqrt{14}}{24} = \boxed{\frac{\sqrt{14}}{6}}$